

Preetika Kaur

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About me: I am a PhD candidate answering the question of where and why L-band InSAR can be trusted to retrieve snow water equivalent. I mapped its feasibility across the western US using GEE, quantifying how forest cover, incidence angle, and snowpack liquid water control coherence using NASA SnowEx UAVSAR data and a distributed snow model, and testing whether those relationships transfer to NISAR. In parallel, I use ground-penetrating radar and variogram analysis from SnowEx 2023 Alaska to characterize snow density variability across tundra and boreal forest. I want to utilize radar observations to create spatially continuous snow information that can feed land surface models and improve streamflow prediction in snow-dominated basins.

Education

Ph.D. in Civil Engineering

2027

University of Wyoming GPA 3.93/4.0

Laramie, WY, USA

Advisor: Dr. Ryan Webb

Dissertation: Advancing understanding of snow remote sensing through L-band InSAR feasibility mapping, coherence controls, and spatial variability across different snow environments

M.S. in Biosystems Engineering

2023

Auburn University GPA 4.0/4.0

Auburn, AL, USA

Advisor: Dr. Jasmeet Lamba

Thesis: Effects of tillage practices and cover crops on X-ray computed tomography derived soil pore characteristics

B.S. in Agricultural Engineering

2020

Punjab Agricultural University

Ludhiana, Punjab, India

Honors, Awards & Scholarships

Research and presentation awards

- 2025 Second Prize, Three-Minute Thesis Competition- University of Wyoming
Intra-collegiate winner and university-wide second place winner
- 2024 Boyd Scott Graduate Student Paper Award – ASABE Annual International Meeting
Second Place Winner, Anaheim, CA
- 2024 Master's Thesis Award- Auburn University Graduate School
- 2024 Outstanding Master's Student Award- Auburn University Graduate School
- 2023 Graduate Student Best Paper Award- AABFEIO, ASABE Annual International Meeting
First Place Winner, Omaha, NE
- 2023 Presentation Excellence Award- ASABE Annual International Meeting
Omaha, NE

Scholarships & fellowships

- 2024, 2025, 2026 International student scholarship- University of Wyoming
- 2025 Michaelis Engineering Scholarship - University of Wyoming
- 2022 100+ Strong Women Abroad, Women in Engineering Scholarship- Auburn University
- 2019-2020 R. N. Kaul Memorial Scholarship - Punjab Agricultural University
- 2016-2019 Merit Scholarship-Punjab Agricultural University

Travel & conference support

- 2026 Travel scholarship, 2026 Collegiate Poster Finalist- WE26 Annual International Conference
Boston, MA
- 2025 Travel scholarship, USC Marine Science Institute- Open Geoscience Hackathon
Marina del Rey, CA
- 2024 Travel and accommodation support, UW eScience Institute- SnowEx Hackweek
Seattle, WA
- 2024 Travel scholarship- AI+Science Summer School, University of Chicago Data Science Institute
Chicago, IL

2024	Travel support, International Arctic Science Committee- Arctic Science Summit Week Edinburgh, UK (online)
2024	Student assistantship- Esri Developer Summit Palm Springs, CA
2024	Graduate student assistantship- Esri Federal GIS Conference Washington, DC
2023	Travel grant, Imageomics Institute-Image Datapalooza Columbus, OH
2023	Graduate student travel award, Auburn GSA, ASABE Annual International Meeting Omaha, NE
2022	Graduate student travel award, Auburn GSA- ASA, CSSA, SSSA International Annual Meeting Baltimore, MD

Publications

First-authored (4)

- Kaur, P.**, Webb, R., Tarricone, J., Rittger, K., McGrath, D., Gagliano, E., Palomaki, R., Bonnel, R., Forster, R., Marshall, H.P., (2026) Feasibility Mapping of L-band InSAR for SWE retrieval in the Western United States. *Geophysical Research Letters*. <https://doi-org.uwyo.idm.oclc.org/10.1029/2025GL120162>
- Kaur, P.**, Lamba, J., Way, T. R., Balkcom, K. S., Sanz-Saez, A., & Watts, D. B (2025) Effects of tillage practices on X-ray computed tomography-derived soil pore networks. *Soil & Tillage Research* <https://doi.org/10.1016/j.still.2024.106035>.
- Kaur, P.**, Lamba, J., Way, T. R., Balkcom, K.S., & Watts, D.B. (2023). Effect of Image Resolution and Soil Core Diameter on Soil Pore Characteristics Quantified Using X-Ray Computed Tomography. *Journal of Soils and Sediments*. <https://doi.org/10.1007/s11368-025-04105-8>
- Kaur, P.**, Lamba, J., Way, T. R., Sandhu, V., Balkcom, K. S., Sanz-Saez, A., & Watts, D. B. (2023). Cover crop effects on X-ray computed tomography-derived soil pore characteristics. *Journal of Soils and Sediments*, <https://doi.org/10.1007/s11368-023-03596-7>

Co-authored (1)

- Sandhu, V., Lamba, J., **Kaur, P.**, Malhotra, K., Way, T.R., Balkcom, K.S., Prasad, R., (2025). Effect of cover crops on phosphorus and trace metal leaching in agricultural soils. *Agric. Water Manag.* 309, 109343. <https://doi.org/10.1016/j.agwat.2025.109343>

Technical Skills

Operating Systems	macOS, Microsoft Windows, Linux
Programming Languages	Python (<i>proficient</i>), R (<i>intermediate</i>), SAS (<i>intermediate</i>), Fortran (<i>intermediate</i>), SQLite, Bash
Snow & hydrologic models	SnowModel (Fortran), HEC-HMS, MODFLOW, SWAT, HYDRUS-1D, RETC, Rosetta pedo transfer (rosetta-soil)
Remote sensing data	UAVSAR (L-band InSAR), Sentinel-1, airborne LiDAR, ground-penetrating radar
Geospatial & analysis tools	Google Earth Engine, ArcGIS, ArcPy, ImageJ, xarray, GeoPandas, Rasterio
Cloud & collaborative computing	CryoCloud, CyVerse Discovery Environment, GitHub, CUAHSI HydroShare
Snow & geophysical field instrumentation	MagnaProbe, WISER probe, Infrasn timer, capacitance plate, ground-penetrating radar, seismic refraction arrays, density wedge cutters, federal samplers, snow pit stratigraphy
Soil & water instrumentation	Campbell data loggers and met station processing, soil moisture sensors, METER ACCUPAR LP-80 PAR/LAI ceptometer, soil augers and core samplers, Trimble Connect GPS, ADC streamflow tracker
Laboratory instrumentation	X-ray CT scanner, muffle furnace, soil hydrometer, peristaltic pumps, bromide probes, pH and EC meters
Machine shop	LeBlond engine lathe, band saw
Human languages	English (fluent), Punjabi (fluent), Hindi (fluent)

Research Work Experience

Graduate Research Assistant

2023 – 2026

University of Wyoming – Department of Civil & Architectural Engineering

Research projects:

- Built an interactive Google Earth Engine tool mapping L-band InSAR SWE-retrieval feasibility across western US mountain ecoregions from forest cover, snow class, and snow wetness
- Quantify how canopy cover, incidence angle, and modeled liquid water content control UAVSAR coherence
- Derive spatially distributed snow density and compare variogram correlation lengths between tundra and boreal sites

Graduate Research Assistant

2021 – 2023

Auburn University – Department of Biosystems Engineering

Research projects:

- Quantified how cover crops reshape soil macropore architecture in strip-tillage cotton fields using high-resolution X-ray computed tomography
- Compared strip-tilled and conventionally tilled soils across two growing seasons
- Established the methodological trade-offs in CT-based pore quantification by testing scan resolution and soil core diameter

Field Work Experience

Seismic refraction surveys- Sugarloaf and Crow Creek, WY; Dry Lake, CO

2026

- Ran refraction lines on paired north- and south-facing slopes to resolve depth to bedrock and its aspect dependence across three watersheds in the Medicine Bow Mountains, Happy Jack, and Steamboat Springs.

Snow hydraulic conductivity method development- Medicine Bow Mountains, WY

2026

- Used the constant-head permeameter method for estimating the hydraulic conductivity of snow.

Stream discharge surveys- No Name watershed, Medicine Bow Mountains, WY

2026

- Measured and processed discharge by conservative salt-tracer dilution gauging.

Canopy structure surveys- Crow Creek, WY and Dry Lake, CO watersheds

2026

- Measured leaf area index with a handheld METER ACCUPAR LP-80 PAR/LAI meter to characterize forest cover for snow and radar analyses.

CSSL Snow Science School- Soda Springs, CA

2025

- Snow pit analysis, thermal imaging, and measurements with spectrometers, liquid water sensors, and federal samplers.

SNOWWI field campaign- Grand Mesa, CO

2024

- Operated Infrastow, WISER probe, capacitance plate, MagnaProbe, and ground-penetrating radar; snow pit density and stratigraphy measurements.

Soil sampling for conservation agriculture trials- E.V. Smith Research Center, AL

2022

- Collected intact soil columns for X-ray CT scanning and auger cores (50 mm diameter, 100 mm deep) at five depth classes across cover crop and tillage treatments.

Laboratory Experience

X-ray computed tomography of soil pore networks- Auburn University

2021

- Saturated intact soil columns and prepared them for CT scanning, then segmented and quantified macroporosity, pore connectivity, and anisotropy in ImageJ and Python across tillage and cover crop treatments.

Soil physical and chemical characterization- Soil and Water Engineering Laboratory, Auburn University

2022

- Measured organic matter by loss on ignition, particle size distribution by hydrometer, pH and EC on soil slurries, and bulk density from oven-dried mass; estimated saturated hydraulic conductivity with the Rosetta pedo transfer model (rosetta-soil Python package).

Rainfall simulation experiments- Soil and Water Engineering Laboratory, Auburn University

2022

- Ran simulations using peristaltic pumps, bromide tracers, and pH and EC instrumentation.

Teaching Experience

Graduate Teaching Assistant, CE3330- Hydraulic Engineering

2026

University of Wyoming

- Lead in-class problem sessions on hydrostatics, pipe flow, hydrodynamics, pump selection, steady and rapidly varied open channel flow, hydrology, and the design of spillways, gates, and culverts.
- Teach four labs: pipe network modeling in EPANET, open channel flow analysis, a hydraulic jump flume experiment, and rainfall-runoff modeling in HEC-HMS.
- Grade homework sets and lab reports and hold office hours for students seeking additional help.

Introduction to Remote Sensing- Lab developer, WyomingView summer internship

2026

University of Wyoming, School of Computing

- Developed an "Introduction to Google Earth Engine" lab sequence for an introductory remote sensing course translating technical concepts into step-by-step workflows for students using GEE for the first time.
- Built exercises covering the GEE public data catalog, spectral index computation (NDVI, NDWI, NBR), and supervised and unsupervised image classification.

Workshop Instructor, Google Earth Engine, ASPRS student organization

2026

University of Wyoming

- Conducted a hands-on GEE workshop introducing cloud-based geospatial analysis and remote sensing workflows.
- Trained participants to use the GEE Code Editor to load, filter, visualize, and interpret Sentinel-2 satellite imagery.
- Demonstrated practical mapping applications, including study-area creation, image collection filtering, cloud masking, seasonal image comparison, and map-layer visualization.

Graduate Teaching Assistant, BSEN 6220- Geospatial Technologies in Biosystems Engineering

2023

Auburn University

- Assisted 57 undergraduate students with GPS (Trimble-connected device) and ArcGIS-related labs
- Conducted grading for student assignments to ensure their understanding of the topic
- Created lab assignments for students related to Python interface within ArcMap

Hackathons, Workshops & Training

Open Geoscience Hackathon- University of Southern California, Marina del Rey, CA

2025

- Built a demonstration Python package end to end to practice open-source packaging workflows, and began developing a personal package for open-source L-band InSAR mapping.

SnowEx Hackweek- eScience Institute, University of Washington, Seattle, WA

2024

- Completed tutorials in geospatial analysis, snow property mapping, remote sensing, and machine learning using Python, xarray, GeoPandas, and Rasterio.
- Contributed to the SnowEx Map Tool project, an interactive tool for visualizing snow properties, collaborating through GitHub on high-resolution spatial data processing.

AI+Science Summer School- Data Science Institute, University of Chicago, Chicago, IL

2024

- Studied AI-driven climate modeling including uncertainty quantification, downscaling, data assimilation, AI emulators, and probabilistic forecasting.

ESIIL Environmental MosAic Hackathon- University of Colorado Boulder (online)

2023

- Built a data cube in the CyVerse Discovery Environment for a machine learning model predicting fire occurrence in Alaska under varying climate conditions.

GeoSMART Hackweek- eScience Institute, University of Washington, Seattle, WA

2023

- Optimized a machine learning model for a snow extrapolation team project in the CryoCloud computing environment.

Image Datapalooza, Imageomics Institute- The Ohio State University, Columbus, OH

2023

- Worked with an interdisciplinary team on land cover data exploration for iNaturalist image locations, alongside Software Carpentry training.

Presentations

Oral presentations

- 2024 Characterization of soil pores using X-ray CT [Boyd Scott Award, 2nd place]
ASABE Annual International Meeting, Anaheim, CA
- 2023 Cover crop effects on X-ray computed tomography-derived soil pore characteristics
ASABE Annual International Meeting, Omaha, NE
- 2022 Effects of cover crops and tillage practices on soil pore characteristics
ASA, CSSA, SSSA International Annual Meeting, Baltimore, MD
- 2022 Effects of image resolution and sample size on soil pore characteristics [lightning talk]
Alabama Water Resources Conference, Orange Beach, AL

Poster presentations

- 2025 Feasibility mapping of L-band InSAR for SWE retrievals
AGU Fall Meeting, New Orleans, LA
- 2023 Temporal and spatial variation of X-ray CT derived soil pore networks under conventional and conservation tillage practices [Presentation Excellence Award]
ASABE Annual International Meeting, Omaha, NE
- 2022 Quantifying effects of soil management practices on X-ray computed tomography derived soil pore properties
Graduate Engineering Research Showcase, Auburn University, AL
- 2022 Effects of cover crops and tillage practices on soil pore characteristics
Alabama Water Resources Conference, Orange Beach, AL
- 2022 Biopore quantification based on macropore shape in pastures
Auburn Research Student Symposium, Auburn University, AL

Service, Leadership & Outreach

Reviewer

Journals: Scientific Reports (Springer Nature)

Conferences: Society of Women Engineers Annual Conference (WE26)- Call for Papers breakout session reviewer, 2026

Leadership and officer roles

- 2026 - 2027 Vice President- Society of Women Engineers (SWE)
University of Wyoming
- 2025 - 2026 Secretary- American Society for Photogrammetry and Remote Sensing (ASPRS) student chapter
University of Wyoming
- 2025 - 2026 Secretary and JEPSC representative- Society of Women Engineers (SWE)
University of Wyoming
- 2025 - 2026 Secretary- Wyomingites in Math, Science and Engineering (WiMSE)
University of Wyoming
- 2024 - 2025 Marketing Head- Society of Women Engineers (SWE)
University of Wyoming
- 2024 - 2025 Cultural Representative- Indian Student Organization
University of Wyoming
- 2022 - 2023 Vice President and member- Alpha Epsilon, honor society for agricultural and biological engineers
Auburn University
- 2022 1887 Graduate Student Ambassador-graduate representative of the Auburn Family and Auburn Creed
Auburn University

Event and program organization

- 2024 Hackweek organizer-SnowEx Hackweek, eScience Institute
University of Washington, May - August 2024
- 2023 Coordinator, 2023 Diwali cultural program- Indian Student Organization
University of Wyoming

Mentorship

2022 - 2023 R. Day- Auburn University, First-Year Graduate Mentorship Program
Mentored an incoming PhD student in Biosystems Engineering

Outreach and volunteering

2022, 2023 E-Day volunteer, Samuel Ginn College of Engineering- represented Biosystems Engineering
Auburn University, March 2022 and March 2023

2022 Outreach research- The Ecology of Opportunity: Environmental Health and Student Learning in Alabama
Auburn University

2022 Gator-Byte water quality testing, Chewacla River-assisted Dr. Eban Bean and Dr. Eve Brantley
Auburn, AL, April 2022

2022 The Big Event- university-wide community service day for Auburn residents
Auburn, AL, February 2022

2022 Parkerson Mill Creek clean-up and stream restoration- AU Water Resources Center
Auburn, AL, February 2022

2022 Tree planting, Kreher Preserve and Nature Center- AU Office of Sustainability
Auburn, AL, January 2022

2021 Gameday recycling- AU Waste Reduction and Recycling Department
Auburn, AL, October 2021

Professional Memberships

American Society of Agricultural and Biological Engineers (ASABE); Association of Agricultural, Biological, and Food Engineers of Indian Origin (AABFEIO); American Society for Photogrammetry and Remote Sensing (ASPRS); Agronomy, Crop and Soil Science Societies of America (ASA, CSSA, SSSA); Alpha Epsilon Honor Society; Society of Women Engineers (SWE), University of Wyoming and Auburn University.